Answer Key & Teacher Notes

Session 1 — Area of Regular Polygons · Session 2 — Composite Figures

HOW TO USE THIS LESSON'S PAGES

one lesson = two sessions

Warm-Up	About 10 minutes; use items 1–4 for a 5-minute check. Key: Unit Warm-Up Key.
Prerequisite Practice A/B	Before the lesson, for students who need the skill the session uses. Key in that folder.
Guided Notes	During the lesson. Students fill in the blanks as you teach; answers below.
Practice Form A	In class, after the lesson. Two pages. The self-check code tells students if they are on track.
Practice Form B	Same skills, new numbers: reteach, retake, small group or absent students.
Exit Ticket	Last 5 minutes. Two per page; cut in half.
Homework Packet	Sheet 1 after Session 1, Sheet 2 after Session 2. Level 1 has the same problems and answers, with a word bank, step frames and larger type for students who need language support.

SESSION 1 · TEACHER NOTES

Area of Regular Polygons

TALK PROMPTS

- How do you know every center triangle has exactly the same area?
- Why is finding one triangle enough to find the area of the whole sign?
- Where does each triangle's height (the apothem) start and end?

PARTNER TASK

Decagon Tabletop. A regular decagon has sides of 6 cm and an apothem of about 9 cm. One partner finds one center triangle (27 cm²); the other multiplies by the number of triangles (270 cm²). Switch roles for the octagon stepping stone.

WATCH FOR

- Finding the area of only one center triangle and stopping.
- Forgetting the ½ in the triangle area (side × apothem × n).
- Using the distance from the center to a vertex instead of the apothem as the height.

SESSION 2 · TEACHER NOTES

Composite Figures

TALK PROMPTS

- When do you add the pieces and when do you subtract? How do you decide?
- How can you split the L-shaped floor a different way and still get the same area?

PARTNER TASK

Tool Shed Side. A shed side is a rectangle 12 ft by 7 ft with a triangle top (base 12 ft, height 4 ft). One partner finds the rectangle (84 ft²), the other the triangle (24 ft²); together decide to add (108 ft²). Then do the banner: 18 in by 12 in with a triangle (base 6 in, height 12 in) cut away: 216 − 36 = 180 in².

WATCH FOR

- Adding the area of a cut-out (window, vent, notch) instead of subtracting it.
- Forgetting the ½ when a piece is a triangle.
- Using the wrong side length for a piece of an L shape (counting an overlap twice).

ITEM ANALYSIS

suggested DOK; judge the reasoning required, not the question label

Sheet	Standard	1	2	3	4	5	6	7	8	9	10	11
S1 · Form A	6.G.A.1	1	1	1	1	1	2	3	2	2	1	1
S1 · Form B	6.G.A.1	1	1	1	1	1	2	3	2	2	1	1
S2 · Form A	6.G.A.1	1	1	1	1	1	2	3	3	2	1	1
S2 · Form B	6.G.A.1	1	1	1	1	1	2	3	3	2	1	1

Items before the last three are Practice; then Find the Error (explain and correct) and two Test Practice items.

GUIDED NOTES

regular polygon: All sides **equal** and all angles equal.

congruent: Exactly the same **size** and shape.

apothem: Distance from the **center** to the middle of a side.

- Lines from the center to each vertex make **congruent** triangles.
- A regular polygon with n sides makes **n** center triangles.
- Area = n × (½ × side × **apothem**).

Example: 5 sides make **5** center triangles. One triangle: ½ × 10 × 7 = **35** cm². All triangles: 5 × 35 = **175** cm². Answer: Area ≈ **175** cm²

YOU TRY

1. One triangle ½ × 4 × 3.5 = 7 in²; 6 × 7 ≈ 42 in²

2. a. 8 b. 10

EXIT TICKET

1. a. 5 b. ½ × 8 × 5 = 20 ft²

2. One triangle ½ × 10 × 8.7 = 43.5 cm²; 6 × 43.5 ≈ 261 cm²

3. **C** (120 in²)

PRACTICE FORM A

1. a. ½ × 10 × 7 = 35 cm² b. ½ × 16 × 11 = 88 in² c. ½ × 7 × 8 = 28 m² d. ½ × 6 × 9 = 27 ft²

2. a. 6 b. ½ × 9 × 7.8 = 35.1 ft² c. 6 × 35.1 ≈ 210.6 ft²

3. One triangle ½ × 16 × 11 = 88 in²; 5 × 88 ≈ 440 in²

4. One triangle ½ × 7 × 8 = 28 cm²; 8 × 28 ≈ 224 cm²

5. One triangle ½ × 24 × 21 = 252 ft²; 6 × 252 ≈ 1,512 ft²

6. One triangle ½ × 18 × 12.4 = 111.6 m²; 5 × 111.6 ≈ 558 m²

7. No. 45 cm² is only one center triangle. The tile has 6 triangles: 6 × 45 = 270 cm².

8. a. 6 × (½ × 8 × 6.9) = 6 × 27.6 = 165.6 cm² b. 25 × 165.6 = 4,140 cm²

9. Lena forgot the ½ in each triangle's area. One triangle is ½ × 22 × 15 = 165 in², so A = 5 × 165 = 825 in².

10. **C** (6)

11. **C** (100 cm²)

Self-check word **POLYGON**

PRACTICE FORM B

1. a. ½ × 12 × 8 = 48 cm² b. ½ × 14 × 9 = 63 in² c. ½ × 5 × 6 = 15 m² d. ½ × 9 × 10 = 45 ft²

2. a. 6 b. ½ × 12 × 10.4 = 62.4 ft² c. 6 × 62.4 ≈ 374.4 ft²

3. One triangle ½ × 14 × 9.6 = 67.2 in²; 5 × 67.2 ≈ 336 in²

4. One triangle ½ × 6 × 7.2 = 21.6 cm²; 8 × 21.6 ≈ 172.8 cm²

5. One triangle ½ × 20 × 17.3 = 173 ft²; 6 × 173 ≈ 1,038 ft²

6. One triangle ½ × 12 × 8.3 = 49.8 m²; 5 × 49.8 ≈ 249 m²

7. No. 28 cm² is only one center triangle. The tile has 6 triangles: 6 × 28 = 168 cm².

8. a. 6 × (½ × 6 × 5.2) = 6 × 15.6 = 93.6 cm² b. 30 × 93.6 = 2,808 cm²

9. Mateo forgot the ½ in each triangle's area. One triangle is ½ × 10 × 12 = 60 in², so A = 8 × 60 = 480 in².

10. **C** (8)

11. **B** (number of sides)

Self-check word **HEXAGON**

GUIDED NOTES

composite figure: A shape made of two or more **simpler** shapes.

decompose: To **break** a shape into smaller shapes you know.

subtract: To **take away** the area of a piece that is cut out.

- Split a composite figure into rectangles, triangles, or other shapes you **know**.
- **Add** the areas of pieces that are joined together.
- **Subtract** the area of a piece that is cut out.

Example: Whole rectangle: 14 × 9 = **126** cm². Cut-out triangle: ½ × 6 × 4 = **12** cm². The triangle is removed, so **subtract**. Answer: 126 − 12 = **114** cm²

YOU TRY

1. Bottom 8 × 3 = 24 m²; top 3 × 4 = 12 m²; total 36 m²

2. a. subtract b. add

EXIT TICKET

1. a. 50 − 6 = 44 ft² b. 24 + 8 = 32 m²

2. Rectangle 10 × 6 = 60 ft²; triangle ½ × 10 × 4 = 20 ft²; total 80 ft²

3. **C** (132 ft²)

PRACTICE FORM A

1. a. add: 52 cm² b. subtract: 84 in² c. subtract: 81 − 9 = 72 ft² d. subtract: 40 − 4 = 36 m²

2. a. 20 × 14 = 280 in² b. ½ × 14 × 5 = 35 in² c. 280 − 35 = 245 in²

3. 13 × 10 = 130; 5 × 11 = 55; total 185 ft²

4. 22 × 10 = 220; 220 − 25 = 195 ft²

5. 24 × 9 = 216; 216 − 18 = 198 ft²

6. Rectangle 26 × 15 = 390 ft²; roof ½ × 26 × 11 = 143 ft²; total 533 ft²

7. No. Deja forgot to subtract the corner. 800 − 100 = 700 in² is left.

8. a. 30 × 20 = 600 m² b. 600 − ½ × 12 × 10 = 600 − 60 = 540 m² c. 540 × 14 = $7,560

9. Omar added the vent, but the vent is not painted, so its area must be subtracted. Painted area = 209 − 25 = 184 ft².

10. **B** (subtract its area)

11. **C** (two or more simpler shapes)

Self-check word **SUBTRACT**

PRACTICE FORM B

1. a. add: 46 cm² b. subtract: 56 in² c. subtract: 64 − 4 = 60 ft² d. subtract: 60 − 9 = 51 m²

2. a. 24 × 16 = 384 in² b. ½ × 16 × 6 = 48 in² c. 384 − 48 = 336 in²

3. 12 × 9 = 108; 4 × 10 = 40; total 148 ft²

4. 18 × 9 = 162; 162 − 16 = 146 ft²

5. 20 × 8 = 160; 160 − 15 = 145 ft²

6. Rectangle 22 × 14 = 308 ft²; roof ½ × 22 × 10 = 110 ft²; total 418 ft²

7. No. Nico forgot to subtract the corner. 720 − 64 = 656 in² is left.

8. a. 28 × 18 = 504 m² b. 504 − ½ × 10 × 8 = 504 − 40 = 464 m² c. 464 × 12 = $5,568

9. Lila added the vent, but the vent is not painted, so its area must be subtracted. Painted area = 170 − 16 = 154 ft².

10. **A** (55 in²)

11. **C** (65 cm²)

Self-check word **COMBINED**

SHEET 1 · AREA OF REGULAR POLYGONS

1. One triangle ½ × 10 × 12.1 = 60.5; 8 × 60.5 ≈ 484 m²
2. Hexagon: P = 24 cm, A ≈ ½ × 24 × 3.5 = 42 cm²; Pentagon: P = 50 ft, A ≈ ½ × 50 × 6.9 = 172.5 ft²
3. a. ≈ ½ × 30 × 4.1 = 61.5 cm² b. ≈ ½ × 60 × 8.7 = 261 ft²
4. Area ≈ 6 × (½ × 6 × 5.2) = 93.6 ft²; cost ≈ 93.6 × $3 = $280.80
5. He found only one of the 5 triangles. Area ≈ 5 × 22 = 110 cm².

SHEET 2 · COMPOSITE FIGURES

1. Bottom 10 × 4 = 40; top 4 × 4 = 16; total 56 m²
2. a. subtract (wall area − window area) b. add (room area + closet area)
3. Top bar 10 × 2 = 20; stem 4 × 4 = 16; total 36 in²
4. Rectangle 12 × 8 = 96; notch base 12 − 3 − 3 = 6, ½ × 6 × 4 = 12; 96 − 12 = 84 cm²
5. 240 − 25 = 215 ft²
6. **A** (50 m²)
7. **C** (46 in²)

October 2026 conceptual additions — teacher key

Session 1 · Area of Regular Polygons

One triangle ≈ 1/2 × 6 × 5.2 = 15.6 cm²; six ≈ 93.6 cm². The apothem is perpendicular to a side and is approximate, so the result is approximate.

Extension: Accept a mathematically valid contrasting example and an explanation that uses the conditions in this session. Teacher anchor: One triangle ≈ 1/2 × 6 × 5.2 = 15.6 cm²; six ≈ 93.6 cm². The apothem is perpendicular to a side and is approximate, so the result is approximate.

Scoring: 1 point for a valid representation/setup, 1 for accurate mathematics and units, 1 for an explanation connecting the method to the task. Accept equivalent reasoning and suitable oral or diagram responses. Do not infer mastery from confidence alone.

Session 2 · Composite Figures

96 − 12 = 84 cm². One partition: 8 × 8 plus 4 × 5 = 64 + 20 = 84. Accept another labeled nonoverlapping partition.

Extension: Accept a mathematically valid contrasting example and an explanation that uses the conditions in this session. Teacher anchor: 96 − 12 = 84 cm². One partition: 8 × 8 plus 4 × 5 = 64 + 20 = 84. Accept another labeled nonoverlapping partition.

Scoring: 1 point for a valid representation/setup, 1 for accurate mathematics and units, 1 for an explanation connecting the method to the task. Accept equivalent reasoning and suitable oral or diagram responses. Do not infer mastery from confidence alone.